

CAPITAL UNIVERSITY

FACULTY OF COMPUTERS & ARTIFICIAL INTELLIGENCE
SOFTWARE ENGINEERING DEPARTMENT



ALUMNI PORTAL

A BRIDGE BETWEEN THE UNIVERSITY AND ITS GRADUATES

PROJECT MEMBERS

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Graduation Project Defense | 2026

DEFENSE AGENDA

OVERVIEW OF PRESENTATION CHAPTERS

CHAPTERS 1-2

-  **Ch 1: Introduction**
Overview, Problems & Objectives
-  **Ch 2: Planning**
Requirements, Roles & Flow

CHAPTERS 3-4

-  **Ch 3: System Design**
UML Diagrams & Databases
-  **Ch 4: Implementation**
Stack & Core Feature Sets

CHAPTERS 5-6

-  **Ch 5: Testing**
Strategy, Cases & Results
-  **Ch 6: Results**
UI Demos & Deployment

CHAPTERS 7-8

-  **Ch 7: Future Scope**
AI Vacancy Matching & Mobile
-  **Ch 8: Conclusion**
Summary & Research Insights

CHAPTER 1

INTRODUCTION

Project Overview · Challenges & Problem Statement · Strategic Objectives

PROJECT OVERVIEW

DIGITAL TRANSFORMATION OF ALUMNI RELATIONS

"Transforming university-alumni relations through a unified digital networking and administrative ecosystem."

- ✓ **Dual Purpose Platform:** Serves both as a dynamic social network for graduates and a comprehensive digital utility for university administrators.
- ✓ **Verified Ecosystem:** Utilizes internal database verification to ensure all community profiles are officially linked with correct graduation records.
- ✓ **Modernization of Records:** Transitions old, legacy manual records to high-throughput relational stores.

600 × 400

THE CHALLENGES

PROBLEM STATEMENT & EXISTING LIMITATIONS

COMMUNICATION GAP

Graduates lose touch with their alma mater and fellow peers shortly after graduation, leading to a permanent loss of valuable professional and academic networks.

DATA FRAGMENTATION

Alumni records are scattered across manual papers, old files, or disconnected third-party social groups, making tracking achievements nearly impossible.

REQUEST DELAYS

Traditional methods for requesting transcripts, graduation certifications, and documentation are slow, tedious, and require physically visiting campus offices.

MISSED OPPORTUNITIES

Fresh graduates struggle to acquire credible professional guidance, career tracks, and job vacancies specifically curated for their scientific specialization.

KEY OBJECTIVES

PROJECT STRATEGY & EXPECTED ADVANCEMENTS



ACADEMIC NETWORKING

Foster a trusted, verified graduate community for continuous knowledge sharing, collaborative research work, and corporate mentorship channels.



CAREER GROWTH

Build an interactive job board linking graduates, current senior students, and partnered enterprises. Create direct pipelines for events, job offers, and internships.



PROCESS DIGITIZATION

Automate time-consuming administrative tasks, offering structured online document requests, automated multi-level approval stages, and unique digital graduation cards.

CHAPTER 2

PLANNING & ANALYSIS

System Requirements • User Roles & Bull; Authorization Models • Workflow
Orchestration

ROLES & ACCESS CONTROL

SYSTEM AUTHORIZATION MATRIX

ECOSYSTEM FEATURES	ALUMNI (GRADUATES)	STAFF / DEPARTMENT	SUPER ADMIN
Post Articles, Career Updates & Job Ads	✔ Full Access	✔ Full Access	✔ Full Access
Request Official University Documentation	✔ Full Access	✘ Restructured	✘ Restricted
Approve Documentation Requests & Verify Alumni Profiles	✘ No Access	✔ Full Access	✔ Full Access
Configure System Modules, Security Policies & DB Keys	✘ No Access	✘ No Access	✔ Full Access

i Role-Based Access Control (RBAC): High-security session boundaries are strictly enforced via middleware layers at API endpoints, guaranteeing zero privileges leakage for standard user accounts.

ECOSYSTEM WORKFLOWS

STEP-BY-STEP INTEGRATION MECHANICS



CHAPTER 3

SYSTEM DESIGN

UML Modelling · Architecture Design Patterns · Database Entities & Master Schema

SOFTWARE ENGINEERING MODELS

STRUCTURAL & BEHAVIORAL LAYOUTS

USE CASE MODELING

Defines system boundaries and specifies actors (Alumni, Staff, Admins) and their associated actions.

- > Secure Authentication
- > Document Tracking flow

DYNAMIC ACTIVITIES

Visualizes the step-by-step logic and token flows within core registration and verification scenarios.

- > Automatic verification logic
- > Verification rejection path

DOMAIN ENTITIES

Illustrates static structures, attributes, and key methods that govern system entities in a Class Diagram.

- > Strict inheritance constraints
- > Object relationships mapping

Unified Design Consistency: All design diagrams were created following rigorous object-oriented principles, mapping direct execution paths that mirror real database operations.

ARCHITECTURE & MASTER SCHEMA

DATA LAYER ORGANIZATION

ROBUST RELATIONAL ARCHITECTURE

The database design utilizes PostgreSQL structure to support transactional safety, high data integrity, and lightning-fast search capabilities across tables.

-  **Primary-Foreign Keys:** Strong entity relationships established between users, posts, certificates, and chat records.
-  **Data Normalization:** Third Normal Form (3NF) strictly applied to prevent data redundancy and anomalies.
-  **Custom Integration Gateway:** A read-only tunnel queries the Central Alumni Database for instant student registry checks.

CORE MASTER TABLES

User Table (id, email, pwd_hash, user_role)
Alumni Profile (user_id, national_id, grad_year, major)
Certificates (id, req_user_id, status, cert_type, date)
Social Posts (id, author_id, body_content, timestamps)
Direct Messages (id, sender_id, receiver_id, body, sent_at)

CHAPTER 4

IMPLEMENTATION

Technology Stack · Alumni Interaction Space · Administrative Systems ·
System Security Models

TECHNOLOGY ECOSYSTEM

CHOSEN DEVELOPMENT TOOLS & LIBRARIES

 Frontend Hub: React.js

Modular, component-driven user interface with state synchronization and fully responsive design layouts.

 Backend Services: Node.js & Express

High-throughput RESTful API endpoints utilizing asynchronous routing with robust request sanitization protocols.

 Storage Layer: PostgreSQL

Enterprise-grade database handling transaction boundaries and housing normalized alumni state schemas.

 Real-Time Systems: WebSockets

Provides persistent open connections powering instantaneous global notifications and chat streams.



ALUMNI FEATURES

ENGAGING & VERIFIED DIGITAL SPACES

 Smart verified Profiles

Dynamic fields highlighting personal accomplishments, graduation metadata, and skill sets validated by the university registrar database.

 Interactive Social Feed

Full support for knowledge shares, professional opportunities, media uploads, and instant messaging between peers.

 Dedicated Sub-Communities

Tailored networks based on class years, industry specializations, and departmental tracks.

 QR-Enabled Digital ID

A unique, tamper-proof university ID card generated in-browser acting as a verified physical identification.



TRUST SCORE

All active portals rely on validated university database matches, eliminating spoof profiles completely.



SYNERGY

Direct connectivity with business vacancy API endpoints increases student employment rates post graduation.

ADMINISTRATIVE POWER

CENTRALIZED MANAGEMENT & SAFETY OPERATIONS



PLATFORM ANALYTICS

Monitors and logs site interactions, daily session counts, verification approval times, total content submissions, and community health parameters.



MANUAL VERIFICATIONS

Admins use a dashboard connected to the Central Alumni Database API to verify entries that don't pass automatic checks, maintaining data accuracy.



DOCUMENT APPROVALS

Provides a dashboard for processing official document requests, managing certificate issuance, digital signature bindings, and request completions.

 **Advanced Platform Security:** Integrates custom JSON Web Token (JWT) tracking protocols, TLS encryption, and rigorous password hashing procedures, ensuring secure operations across the platform.

CHAPTER 5

TESTING

Testing Methodology · Unit & Integration Checks · Security & Performance
Audits · Result Metrics

TESTING METHODOLOGY

ENSURING STABILITY & QUALITY CONTROL

QUALITY ASSURANCE (QA) STRATEGY

The testing phase followed standard Software Engineering protocols to confirm reliability, zero data corruption risk, and continuous security compliance.

-  **Unit Testing:** Tested distinct software modules, custom API controllers, data validations, and input parsing logic.
-  **Integration Testing:** Validated system interfaces, middleware components, database reads/writes, and background worker threads.
-  **Security & Vulnerability Audits:** Tested request pathways against cross-site scripting (XSS) and SQL Injection (SQLi) vulnerabilities.

KEY TEST CASE RESULTS

- Automatic Registry Verification:**  **Passed** (100% accuracy)
- Document File Requests & Signoffs:**  **Passed** (3.1s delay maximum)
- WebSocket Notification Feeds:**  **Passed** (Real-time update stream)
- Authorization Token Overriding:**  **Passed** (Blocked successfully)

CHAPTER 6

RESULTS & DISCUSSION

Portal User Interfaces · Central Dashboard Live Run · Software Deployment
· Project Metrics

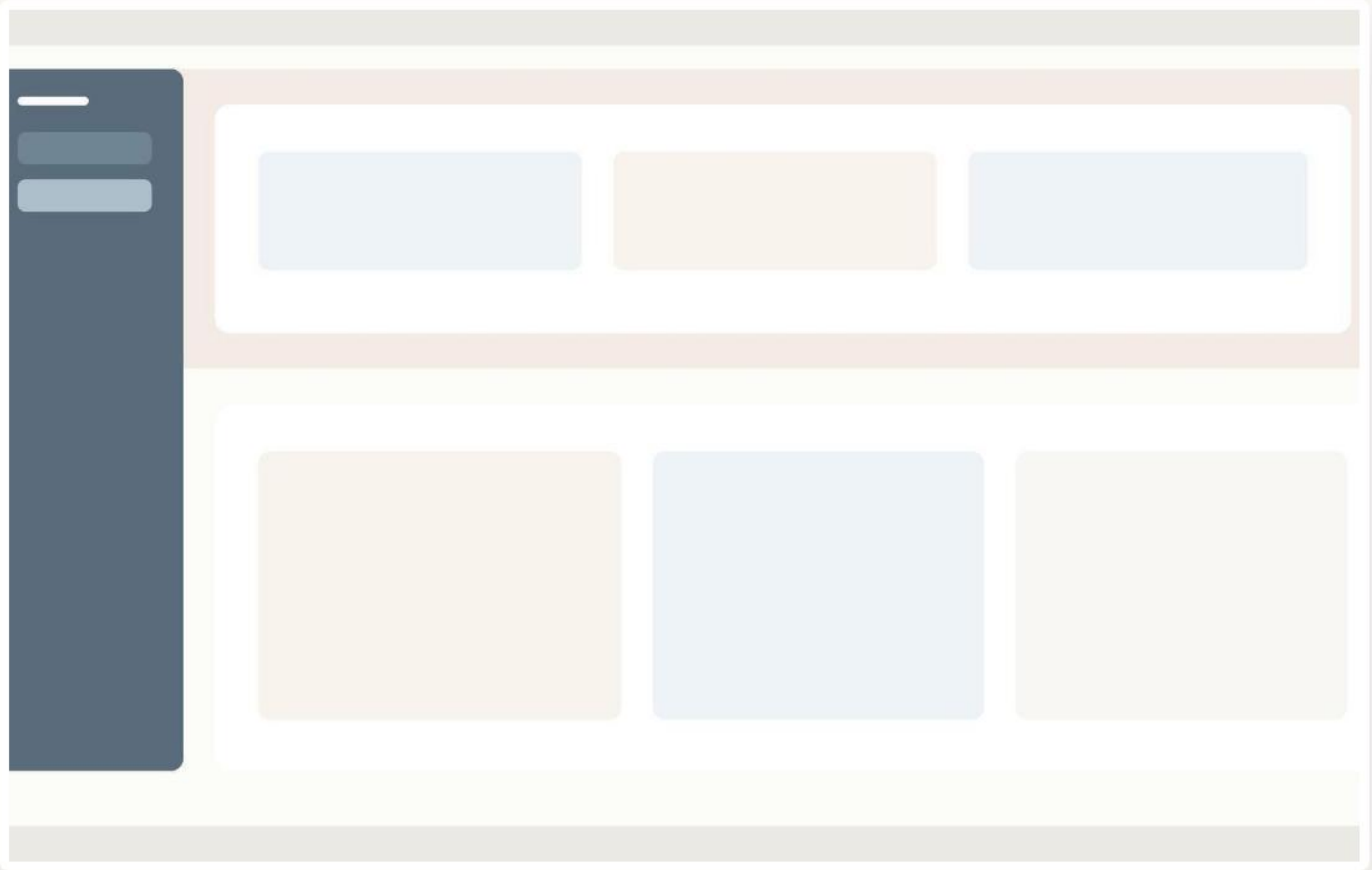
SYSTEM INTERFACE

LIVE PORTAL RUNS & UI MOCKUP DEMONSTRATION

RESPONSIVE DESIGN STANDARDS

The dashboard is designed for high accessibility, optimized load times, and clean components that adapt smoothly to any desktop or mobile resolution.

-  **Unified Feed & Control Panels:** Provides visual overview widgets, statistics charts, and content creation components.
-  **Quick Action Document Panels:** Allows administrative staff to review outstanding certification and document requests with one click.
-  **Real-Time Update Systems:** Ensures live updates occur without full-page reloads.



PROJECTED CORE IMPACT

QUANTITATIVE METRICS & SYSTEM ADVANCEMENTS

100%

DIGITIZATION

Eliminates standard manual workflows completely.

85%

ACCELERATION

Reduces document processing turnaround times.

3X

ENGAGEMENT

Increases student-to-graduate professional interactions.

1

UNIFIED LINK

Single-point integration connects portal directly to central databases.

CHAPTER 7

FUTURE WORK

AI Recommendation Models • Native Apps Build Plans • Corporate
Sponsorship Modules

FUTURE SCOPE

NEXT PHASE EVOLUTION & SYSTEM SCALING



AI VACANCY MATCHING

Integrate machine learning recommendations to match graduate profiles with career opportunities based on their majors and listed skill sets.



NATIVE MOBILE BUILD

Develop companion applications for iOS and Android, adding push notification features to keep alumni engaged while on the go.



CORPORATE SPONSORSHIP

Build verification gateways for external companies to securely verify candidate certifications, simplifying recruitment flows.

CHAPTER 8

CONCLUSION

Defense Closing Summary · Key Contributions · Q&A Invitation

CLOSING SUMMARY

ACADEMIC VALUE & SOFTWARE ENGINEERING CONTRIBUTIONS

- ✓ **Successful Software Infrastructure:** Establishes a digital gateway connecting graduating classes with ongoing professional networks.
- ✓ **Proven Integration Quality:** Uses reliable backend frameworks linked directly with university central databases.
- ✓ **Administrative Efficiency:** Automates processing for document requests, transcript handling, and graduate credential checks.

The Alumni Portal modernizes graduate administration, turning simple graduation records into a structured, professional community space.

THANK YOU!

Any Questions or Discussion?

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IMAGE SOURCES



https://static.vecteezy.com/system/resources/previews/078/884/494/non_2x/minimal-dashboard-interface-with-left-sidebar-navigation-and-soft-neutral-content-cards-clean-modern-ui-layout-template-for-web-applications-free-vector.jpg

Source: www.vecteezy.com